

LDP_internship_report

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Introduction

Timeline

Week 1 (November 3-9th, 2025)

Meeting with David Hunt (Wed 5th)

- > Folder structure - separate intermediate output from final, cleaned output
- > If possible, use SAS files instead of looping the lines of the .dat files
- > I started working with the .dat files because there were missing informations in the SAS data files. Howard sent the SAS metadata on Fri, 7th, then I started working on them.
- > Ask Howard -> Should SAS files and .dat have the same info? HS - Yes, they should, but they haven't.
- > David mentioned there probably are floating point error in some of the columns
- > Add unique IDs for trawls and fishes - trawl unique ID - date, lake_code, trawl_number - fish unique ID - date, lake_code, trawl_number, fish_id

Week 2 (November 10-16th, 2025)

- > Columns trawl_location, sample_type and sample_number are not present in the SAS files - Howard said sample_type and sample_number are probably same for all rows - Athena expressed concerns about the trawl_location, and I will try to maintain the info in the final matrix
- > We will remove the processor/observer names because of privacy purpose - add the justification to the data_dictionary
- > NA should be left as is
- > Round numbers to two decimals after point suring quality check period.
- > There are more rows in the SAS data file than in the .dat file. for year '84
- > "comment" column appears to be empty, if that is the case for all years, it should be removed
- > Create a summary table - compare SAS and .dat files - survey by day (e.g. lake 3, date - compare info)
- recommended match and merge - cross check lake in every year - summarize command - count # lakes in that year - Count # of rows per lake, date, species and trawl.
- > Apple data cross checking confirmation for '77-99 - check if there are original files to all years - looks like we don't

Meeting with David Hunt, Athena Ogden, Howard Stiff (Wed 12th)

Week 3 (November 17-23th, 2025)

-> SAS files “trawl85.sas7bdat”, “trawl86.sas7bdat”, “trawl87.sas7bdat”, “trawl94.sas7bdat”, “trawl97.sas7bdat”, “trawl98.sas7bdat” Contain wrongly formatted information in the metadata. It jeopardized the table merging.

-> Yulia’s script worked well for .dat files from ’84-’86. However, I had to adapt the script to the other years because the loop was assigning values to the wrong columns. For the year ’88, I modified the “start_end_time” cleaning script, and for the ’87 and ’90-’99, I let the empty lines to county as missing data and modified the loop to account for it.

-> Create a summary scrip to compare SAS and DAT files - some of the .dat have still issues with the loop - ’89

Meeting with David Hunt, Athena Ogden, Howard Stiff (Mon 17th)

-> Next steps after the (Mon 17th) meeting: - check duplicates - correct loop year ’89 - How many records have missing duration? Calculate the duration for the missing ones - Commit changes as branch - Assemble all datasets into one giant table - Separate juvenile/adults - Add line number to source files (.dat) - clean name of source file (original name without path) - check with inventory Howard uploaded to gitHub - Send corrupted SAS files to David decode - Compile missing info in these corrupted files and send to Athena and Howard

Week 4 (November 24-30th, 2025)

-> Steps performed after the (Mon 17th) meeting: - corrected loop year ’89 - adapted script 88-99 - Fixed problems with ’87 (three and four lines mixed, spaces between metadata) - separate script to it - .dat ’86 has different time format for different data blocks. Also, Missing metadata info for samples in lines 1253 - 1962 (blocks of empty lines). - Commit changes as branch - Assemble all datasets into one giant table - Add line number to source files (.dat) - clean name of source file (original name without path) - check with inventory Howard uploaded to gitHub - Send corrupted SAS files to David decode - year ’98 dat - last block of data is not being added (?) - fixed by adding a new line at the end of .dat file - reverse order of name and code in year ’88 dat file for depth_m

-> Issues found: - preservative code 11 (98-99), there is no info on weight conversion or the name of it, formalin 0% ? - “0 0% FORMALIN - WGHD AFT 5 WKS” - written as a comment in the preservative column - can I add this in the preservative lookup table? - ’99 “PROCESSED BY ALAN HIRST. COMPUTER WILL NOT ACCEPT 00 FOR DATE OF PROCESSING SO I’AM USING 99.” - ’84 .dat file with huge amount of missing data when compared to equivalent SAS file. - IMPORTANT: Trawl number SAS file ’91 is “999” while it is “99” in equivalent .dat file!!!

-> Steps to be performed during quality check phase: - Compile missing info in these corrupted files and send to Athena and Howard - Check duplicates - How many records have missing duration? Calculate the duration for the missing ones - Separate juvenile/adults

-> Meeting with David Hunt, Athena Ogden, Howard Stiff (Wed 26th) - Use fish unique ID to combine datasets - use “update_values” command of full_join - joyn package - Formalin info should be different depending on the concetration. Add info to different columns - Check how many preservative codes there are in the data (unique()) to correct the “11” in the ’98-99 data files - y2k bug - convert ’99 to 2000 in the processing date of ’99 file and flag it in the metadata and in the data frame. - Add to the metadata that there were data loss in .dat ’84, not only in ’83 and before. - Try standardizing columns and using cbind to concatenate same year data. Use fishID to do it and exclude duplicates later, ignoring source file column. - I had an issue with the pushing/pulling things from GitHub. David will help me later with this.

Week 5 (December 01-07th, 2025)

-> to do: - Seems like there are duplicates in fish unique ID. Maybe add species code to the fishID - Fish total in SAS files has weird format, they are not integers. Check the original file and make it an integer. - y2k bug - convert '99 to 2000 in the processing date of '99 file

-> Issues found: - When I was converting the fish total in SAS file, I noticed that SAS '84 presented different numbers of "fish total" and "fish_id" in the metadata and trawl file. There are more fishes in the trawl file than the number reported in the metadata. e.g. sample date 1984-05-29 - even with the species code in the fish unique ID, the codes are not unique.

-> Meeting with David Hunt, Athena Ogden, Howard Stiff (Mon 01st) - We discussed the possible ways of making the fish unique ID actually unique - Fish ID is an arbitrary number, and blocks of data with the same metadata, but separated in the data files, will be displayed as duplicates. - Use fish length and weight in the ID instead of the fish id - Check duplicates before combining the databases

Week 6 (December 08-14th, 2025)

-> Done: - Removed duplicates in the individual dataframes - Corrected y2k bug in the '99 trawl dat file - Combined data frames keeping rows as duplicates - Fish unique ID is: TrawlDate_LakeCode_TrawlNumber_Depth_SpeciesCode_FishID_Weight_Length

-> Meeting with David Hunt, Athena Ogden, Howard Stiff (Wed 10th) - David was going to try to find a simpler way of combining the data frames because the "full_join" function was not combining all columns and rows. It combined on columns or rows.

Week 7 (December 15-21th, 2025)

-> Done: - Collapsed duplicated rows keeping the .dat info when conflict existed - Flagged problematic time rows across start_time columns in .dat files (column "dat_invalid_start_time") - Collapse source_file columns into one

-> Needs to be done: - Relocate columns - How many records have missing duration? Calculate the duration for the missing ones - Separate juvenile/adults - Check how many preservative codes there are in the data (unique()) to correct the "11" in the '98-99 data files

-> Meeting with David Hunt, Athena Ogden, Howard Stiff (Wed 17th) - Midnight should be 00 or 24 - Return to the .dat file script because there is an issue when converting the format of the time - Have one script calling other scripts - Create a 03_errors folder on GitHub and upload .csv files with the flagged errors - duplicates in the final combined data frame - flag which species have different "fish_total" between .sas and .dat files. - Errors with the start_time, end_time and duration_mi

-> Issues: - fish_unique_ID == "1986-04-16_161_4_15_1_4_3.73_68" - record present in both trawl85.dat and trawl86.dat. The current code keep only the first time the record appears, deleting the second row. So, the Trawl86.dat record is not kept (scale_book is different between these files). Should I keep it like that, as they seem to be duplicates, or should I keep them separately? - For the trawl92.sas file, the "100:MM:SS" is in the original file. It is absent in the .dat file.

Week 8 (December 22-28th, 2025)

-> Done: - Return to the .dat file script because there is an issue when converting the format of the time - The issue was when re-opening the table as csv. I removed that step and saved as csv only at the end. - 10247 rows with wrong format. - Relocate columns - Separate juvenile/adults in fish_description - Check how many preservative codes there are in the data (unique()) to correct the "11" in the '98-99 data files

Week 9 (December 29, 2025 - January 4th, 2026)

-> Done: - How many records have missing duration? Calculate the duration for the missing ones - Errors with the start_time, end_time and duration_mi

-> Issues: - Processor "99" should be replaced by "NA"? Replace all "999" by "NA" - preservative codes "9" and "98" mean "unknown", should I keep both? all become NA - preservative code 11 in trawl 96 = "0 0% FORMALIN - WGHD AFT 5 WKS" - does this mean "fresh"? - preservative code 350 = 0 0% OH - WGHD AFT 5 WKS - preservative code 270 = 0 0 % OH - WGHD AFTER 5 WKS - fish_unique_ID == 1997-09-17_69_9_15_7_1_1.35_49 - duration_min, is the "0203" = end_time? - correct the end time and calculate the duration - fish_unique_ID == "1986-04-16_161_4_15_1_4_3.73_68" - record present in both trawl85.dat and trawl86.dat. The current code keep only the first time the record appears, deleting the second row. So, the Trawl86.dat record is not kept (scale_book is different between these files). Should I keep it like that, as they seem to be duplicates, or should I keep them separately?

-> Needs to be done: - Combine start_time and end_time columns - change order .dat and .sas - Replace all "999" or "99" by "NA" in all columns - correct the end time and calculate the duration - everything with 0203 - fish_unique_ID == 1997-09-17_69_9_15_7_1_1.35_49 - 11 -> 1 - preservative code - add a column duration_mi_comment - collapse all comments columns into one. e.g.: Duration_min: "changed from 1500?, PR 23Mar2000". include "trawl_location" - move Ethanol 95% as preservative_comment but keep the comment - report the change in a general comment column - check the frequency - info from trawl_location column - preservative code 350 = 0 0% OH - WGHD AFT 5 WKS - code 3 (typo error) - preservative code 270 = 0 0 % OH - WGHD AFTER 5 WKS - code 2 (typo error) - season - April 1st to March 31st - Check the books and season times - Wrong hours format that are also wrong in the original .sas and .dat files replace them by NA - Two columns for species: genera and species - standardize species_code_comment column - export the rows with no fish_description - if between FRY and Juvenile - keep Fry